

Project Title

ZeroWaste Connect – Smart Surplus Donation Platform

1. Problem Statement

Large amounts of surplus food and essential resources are wasted daily while NGOs, orphanages, shelters, and old age homes struggle to meet basic needs. Existing donation methods are mostly manual and lack an efficient way to connect donors with recipients.

ZeroWaste Connect provides a centralized platform where donors can donate food, groceries, clothes, money, and other essentials, while recipients can browse donations or post their requirements.

2. Project Objective

- Reduce food and resource wastage.
 - Connect donors with charitable organizations.
 - Support multiple donation types.
 - Simplify donation requests and management.
 - Enable pickup and delivery coordination.
 - Provide a scalable platform for future AI integration.
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3. Proposed Solution

ZeroWaste Connect is a web application that connects donors and recipients through a digital platform. Donors can publish available donations, while recipients can browse donations or create need requests. The application uses **OpenStreetMap + Leaflet** for location management and **Spring Boot Mail** for email notifications, making the donation process simple and transparent.

4. Technology Stack

Layer	Technology
Frontend	React.js
Backend	Spring Boot
Database	PostgreSQL
Security	Spring Security + JWT
ORM	Spring Data JPA
Maps API	OpenStreetMap + Leaflet
Email API	Spring Boot Mail
API Testing	Postman
Build Tool	Maven
Version Control	Git & GitHub

5. Functional Modules

Module 1 – Authentication

- Donor Login
- Recipient Login
- Admin Login
- JWT Authentication

Module 2 – Donor Management

- Add Donation
- Edit/Delete Donation
- View Requests
- Donation History

Module 3 – Recipient Management

- Browse Donations
- Request Donation
- Create Need Request
- View History

Module 4 – Donation Management

- Food
- Groceries
- Clothes
- Money
- Other Essentials

Module 5 – Location Management

- Select Pickup Location
- View Donation Location
- Navigation

Module 6 – Email Notification

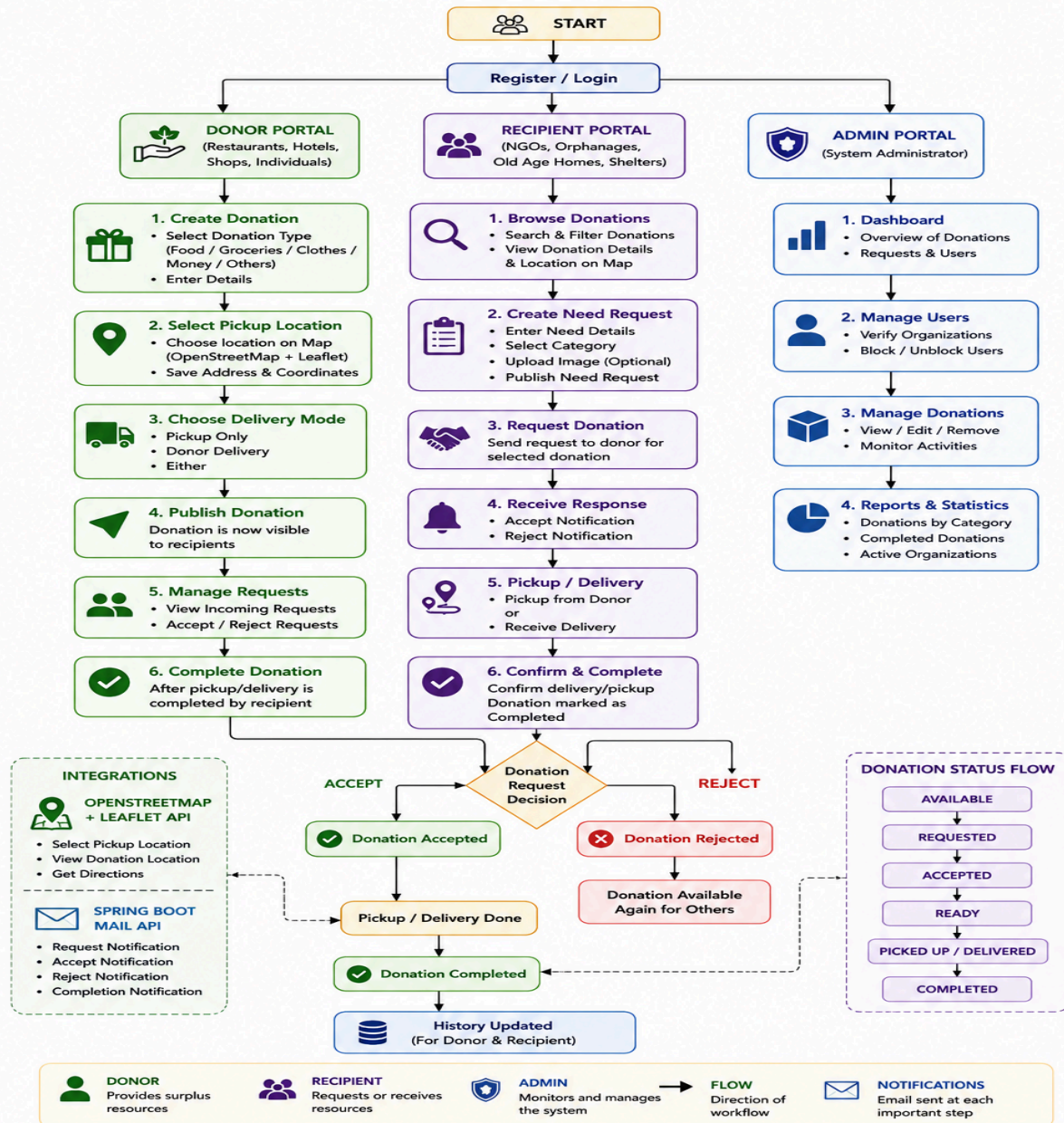
- Request Notification
- Acceptance Notification
- Completion Notification

Module 7 – Dashboard & Admin

- Manage Users
- Manage Donations
- Reports
- Statistics

6. Project Workflow

ZEROWASTE CONNECT – PROJECT WORKFLOW



7. Core Features

- JWT Authentication
- Multi-Resource Donation
- Need Request Management

- Donation Request & Approval
 - Pickup & Delivery
 - OpenStreetMap Integration
 - Email Notifications
 - Dashboard & Reports
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8. External APIs

OpenStreetMap + Leaflet

Used to display and manage donation locations.

Spring Boot Mail

Used to send email notifications for donation activities.

9. Expected Outcome

ZeroWaste Connect provides a simple and efficient platform for connecting donors with organizations in need. It reduces resource wastage, improves donation management, and demonstrates modern full-stack development using React.js, Spring Boot, PostgreSQL, JWT Authentication, REST APIs, OpenStreetMap + Leaflet, and Spring Boot Mail. It also provides a strong foundation for future AI-based enhancements.